

Class 11: Cascades and fads in cultural markets

Matthew J. Salganik

Sociology 204: Social Networks
Princeton University

Monday, March 3, 2025



Review:

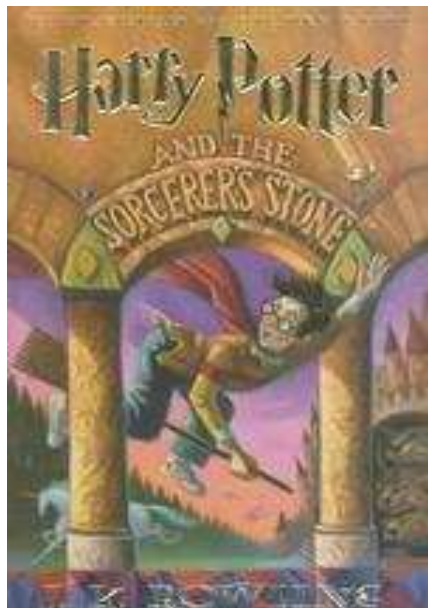
- ▶ sometimes small shocks get big and sometimes they don't

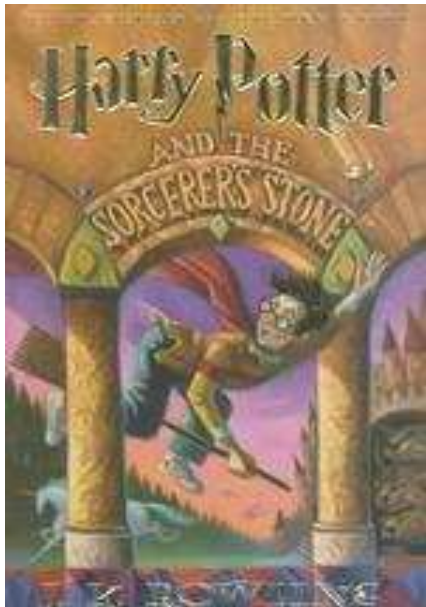
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- ▶ sequential behavior can lead to unexpected outcomes
- ▶ We got to hear from a surprise visitor (Alex) how thresholds are used in real research today





- ▶ Wild success
- ▶ Rejected by eight publishers

This seems like a strange combination.



- ▶ Set box office records, won 6 Oscars, and launched a multi-billion dollar franchise
- ▶ Rejected by United Artists and Universal before being made by Fox



- ▶ One of the most popular shows of the decade
- ▶ Rejected by ABC, CBS, and NBC before being picked up by Fox

Puzzling nature of success for cultural objects (books, movies, piece of art, music)

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- ▶ extreme inequality in the success of objects
- ▶ unpredictability in the success of objects

Psychological explanation:

People agree on what's good, but people are hard to predict

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Sociological explanation:

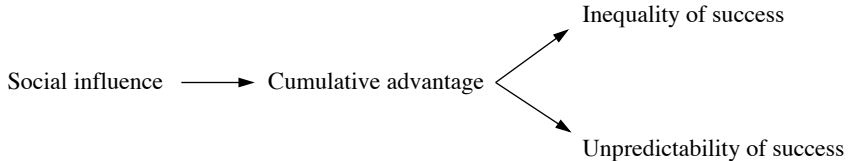
The collective outcomes of inequality and unpredictability of success both arise from an individual-level process of social influence

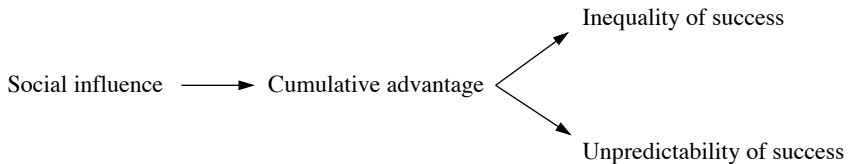
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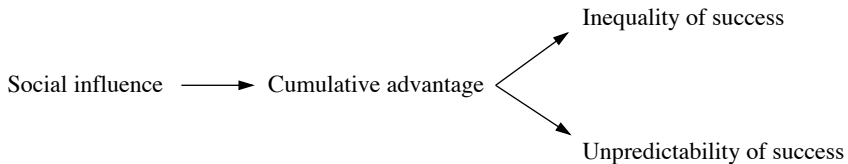
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Problems with observational data:

- ▶ don't know what would have happened without social influence



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- ▶ can't see multiple "histories" to observe unpredictability

Instead of using observational data we are going to run an experiment because

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- ▶ can control the information that people have about the behavior of others
- ▶ can run the same process multiple times under exactly the same conditions, allows us to see multiple “histories”

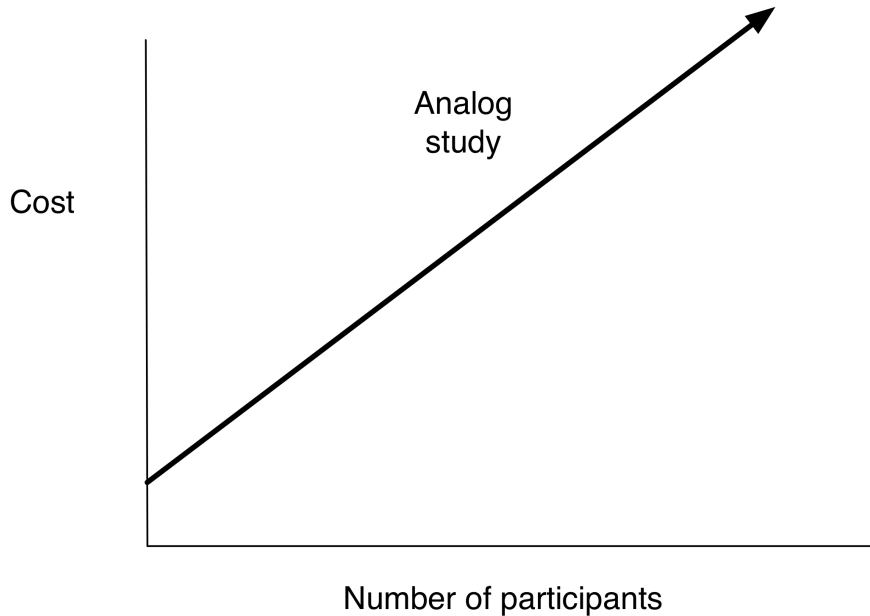
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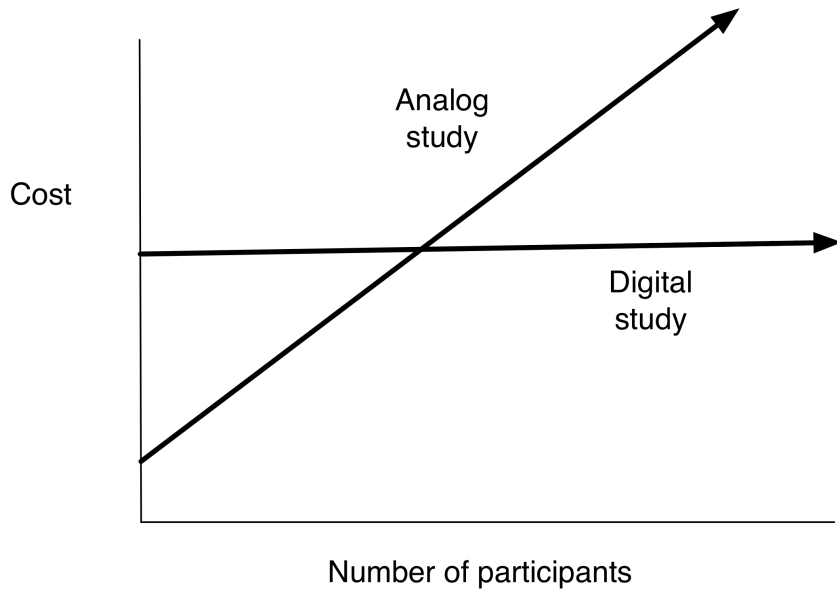
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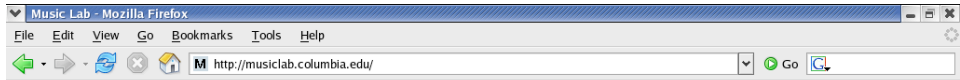
But, this experiment is different from most,

- ▶ experiments in psychology and economics have **individual** as unit of analysis, require **hundreds** of participants
- ▶ these sociological experiments have **collective outcome** as unit of analysis, require **thousands** of participants

Web-based experiment allow for such large sample sizes because each additional participant has no cost (total $n = 27,267$)







 **MUSIC LAB**

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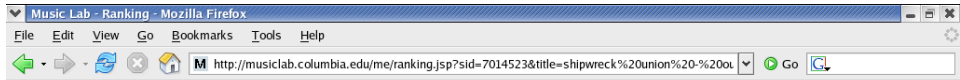
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HARTSFIELD: "enough is enough"	20	GO MOREDCAI: "it does what its told"	12	UNDO: "while the world passes"	24
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00:21 shipwreck union - out of the woods volume

|| ▶

Please rate this song.
You don't need to wait until it's finished

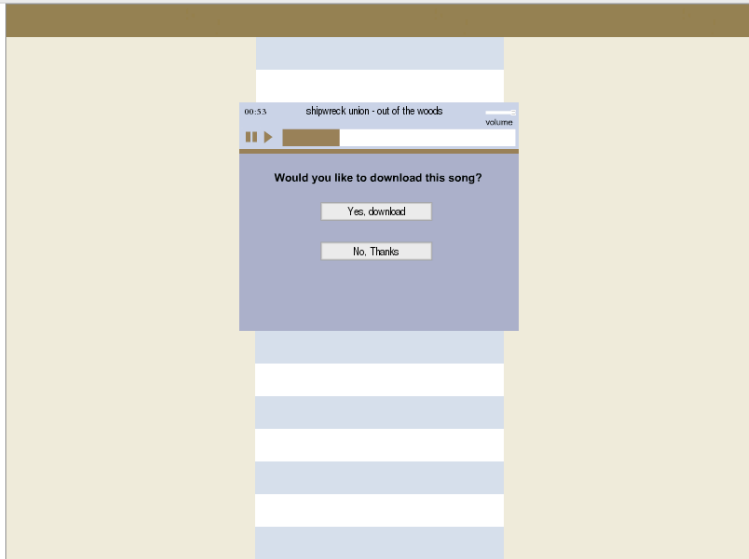
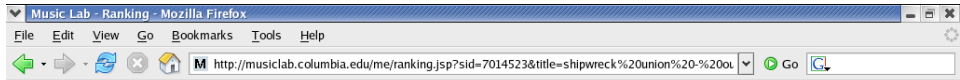
☐ ★ ★ ★ ★ ★ i love it

☐ ★ ★ ★ ★ ★ i like it

☐ ★ ★ ★ ★ it's OK

☐ ★ ★ ★ i don't like it

☐ ★ ★ i hate it



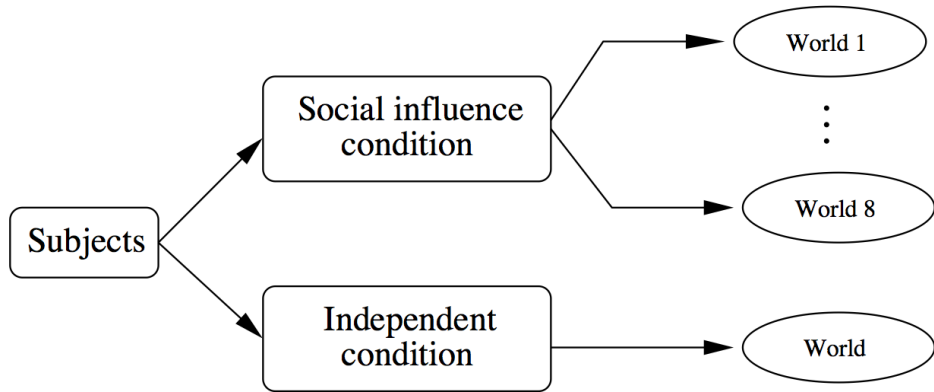
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HARTSFIELD "enough is enough"	20	GO MEDICAL "i don't want to die"	12	UNDO "while the world passes"	24
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THE BROKEN PROMISE: "the end is the end"	19	POST BREAK TRAGEDY: "Riviera"	14	STRANGER: "too deep"	10
THIS NEW DAWN: "the hotel where the answer"	12	FORTIFYING: "you"	24	FAIR FROM KNOWN: "twice 2"	18
MOONER AT RISE: "walk away"	6	THE CALLERATION: "trapped in an orange post"	20	STUNT MONKEY: "inside out"	46
MORAL HAZARD: "waste of my life"	8	CINETRO: "the future"	17	DANTE: "this moment"	14
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AIT OF KAILY: "before the rain, make breakfast"	30	THE PASTLARE: "i don't do as part (i don't)"	31	BY NOVEMBER: "i could take you"	20
HYDRAULIC SANDWICH: "expensive anxiety"	20	A BLINDING SILENCE: "memories and mistakes"	17	DRINKIN IN THE SKY: "tap the side"	12
EMBER SIX: "the upcoming winter"	25	SLIM FANAL: "rain off the city"	15	SELISUS: "rain off the city"	22
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RYAN ELSBARGER: "don't be sad"	14	SP FALLS DOWN: "a broken burning out"	11	EVAN GOLD: "what diversity"	38
DETERIORING: "father to son"	10	SUMMERKAWASTED: "a plan behind destruction"	17	BENEFIT OF A DOUBT: "van away"	10
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(a) Experiment 1, Weaker signal

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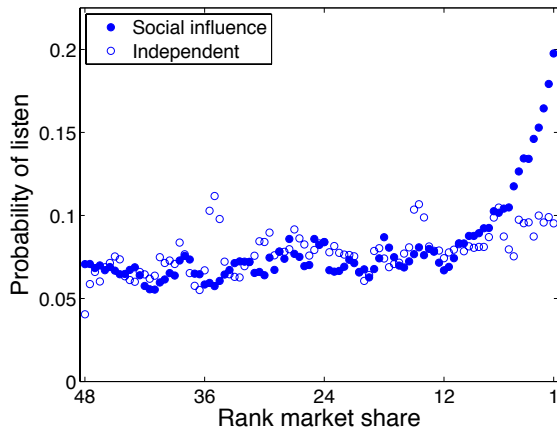
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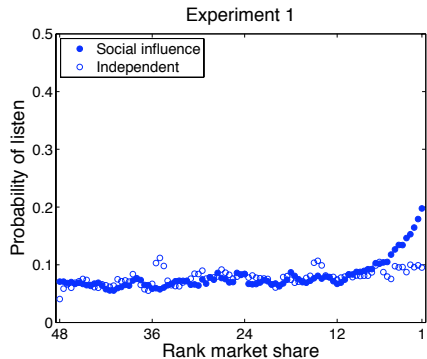
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PARKER THEORY: "the rain"		159
THE PASTLARE: "i don't do as part (i don't)"		103
SELISUS: "rain off the city"		62
STUNT MONKEY: "inside out"		56
BY NOVEMBER: "i could take you"		55
FORTHEDING: "you"		49
HYDRAULIC SANDWICH: "expensive anxiety"		43
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Done

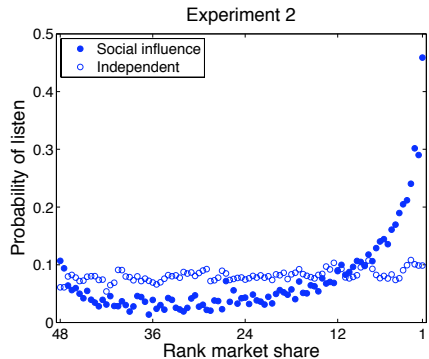
(b) Experiment 2, Stronger signal

Experiment 1



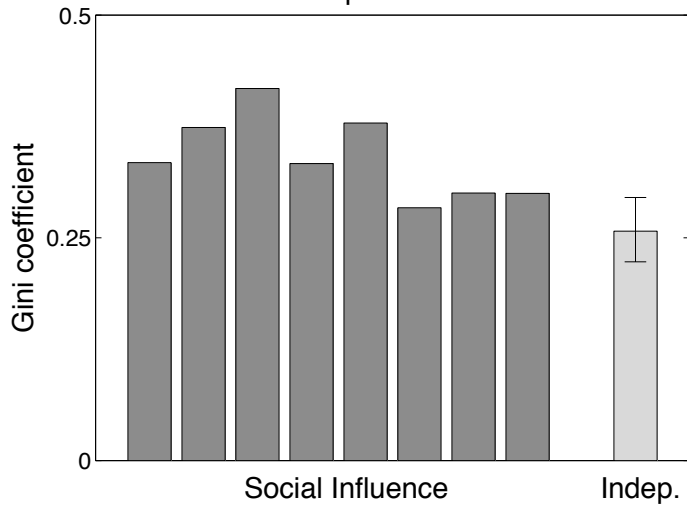


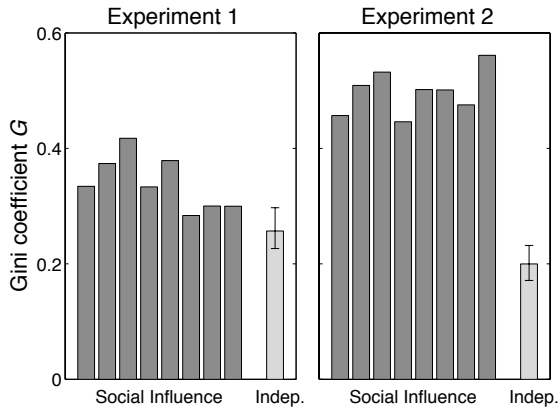
(a) Experiment 1, weaker signal



(b) Experiment 2, stronger signal

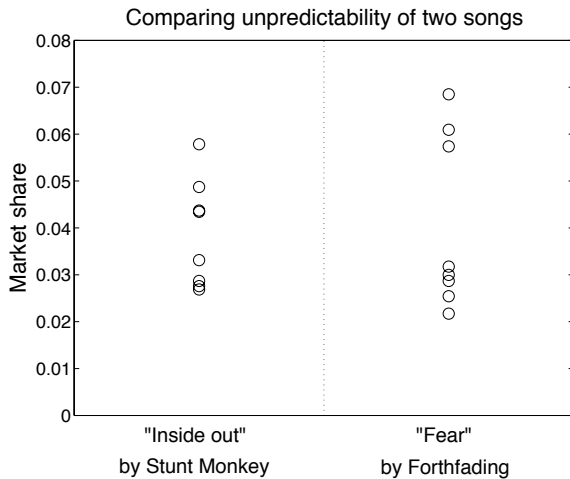
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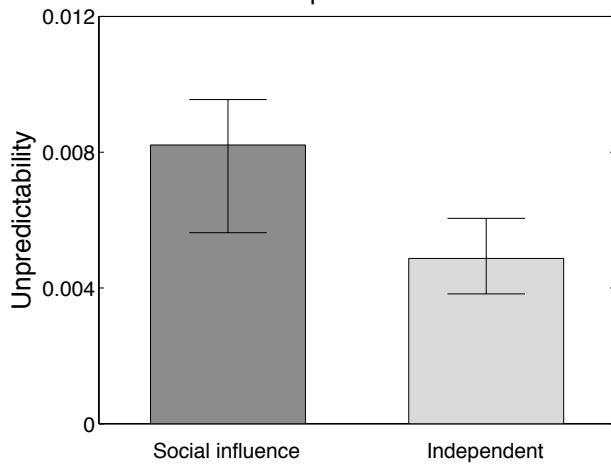


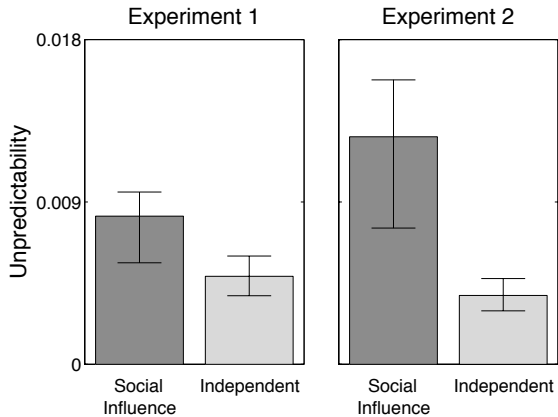
Median Gini coefficient increases from 0.34 (France) to 0.50 (Nigeria)

U = mean difference in market share across all pairs of realizations



Experiment 1



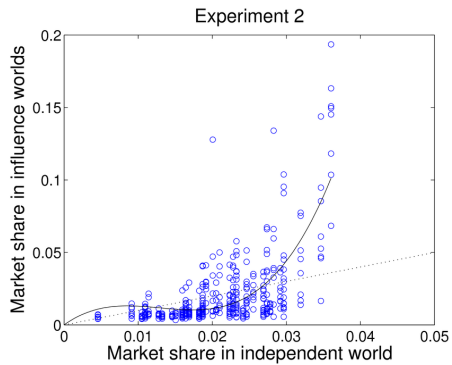
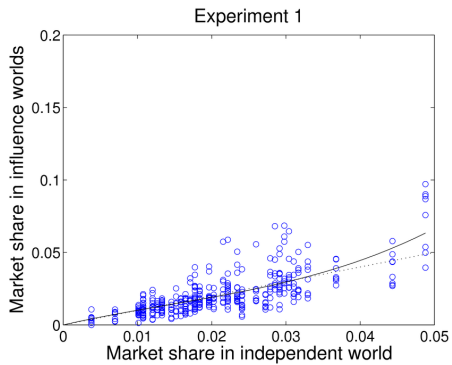


Unpredictability increases by about 50%

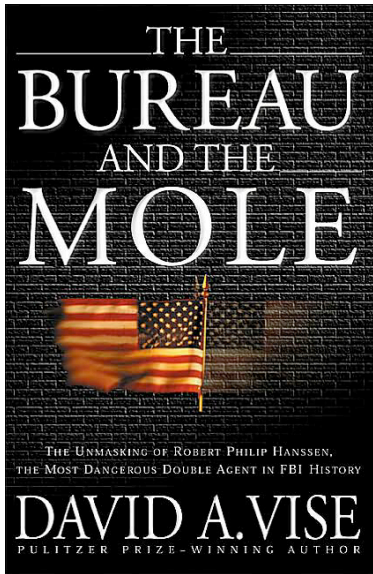
Experiments 1 and 2 show a dose-response relationship. Increasing the strength of social influence leads to

- ▶ increased inequality of success
- ▶ increased unpredictability of success

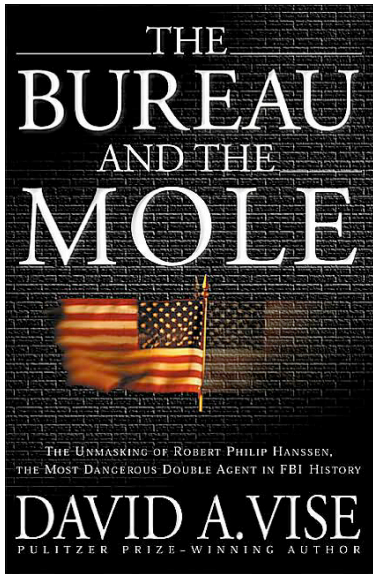
What is the relationship between “quality” and success?



If real success can lead to more success, can fake success lead to real success?



Someone bought and returned 17,000 copies of this book.



Someone bought and returned 17,000 copies of this book. It was David Vise.

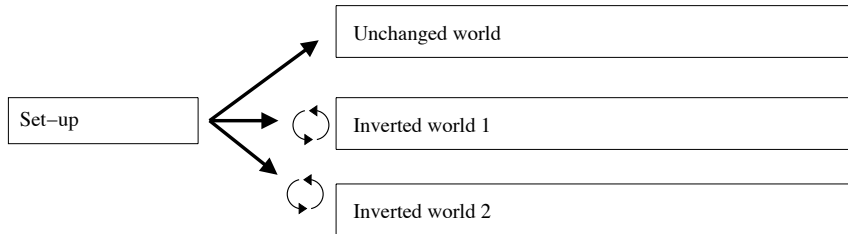
Self-fulfilling prophecies

“[a] self-fulfilling prophecy is, in the beginning, a *false* definition of the situation evoking a new behavior which make the originally false conception come *true*.” Robert Merton (1948)

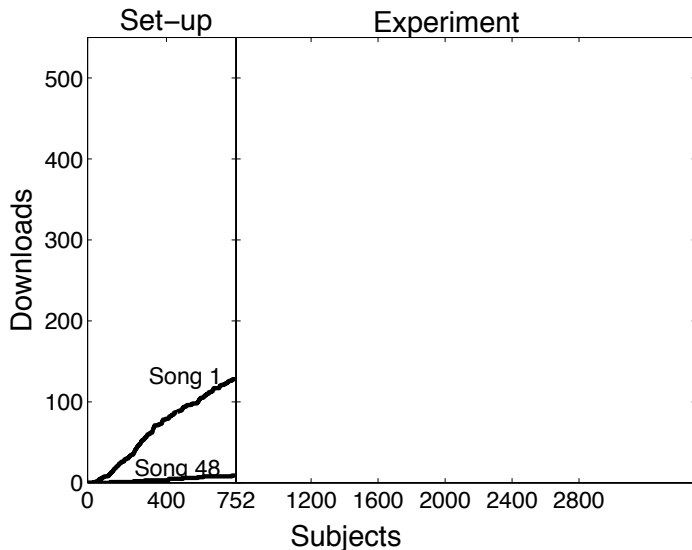
Vise's book sold more than 180,000 copies. How much of this was due to a self-fulfilling prophecy? Hard to say.

Again, this is difficult with observational data, but possible with a multiple-realization experiment

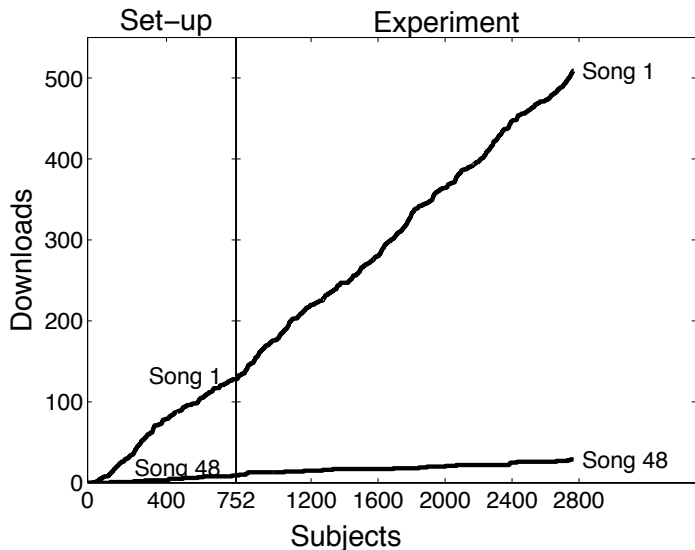
Experimental design



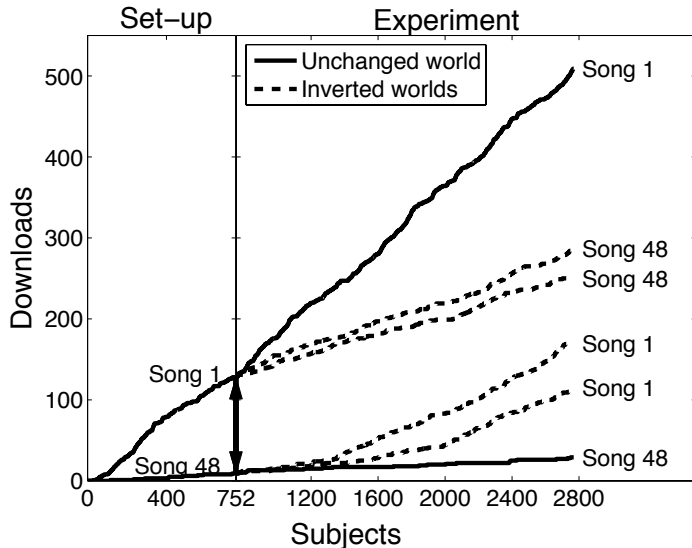
Tracking song 1 and song 48



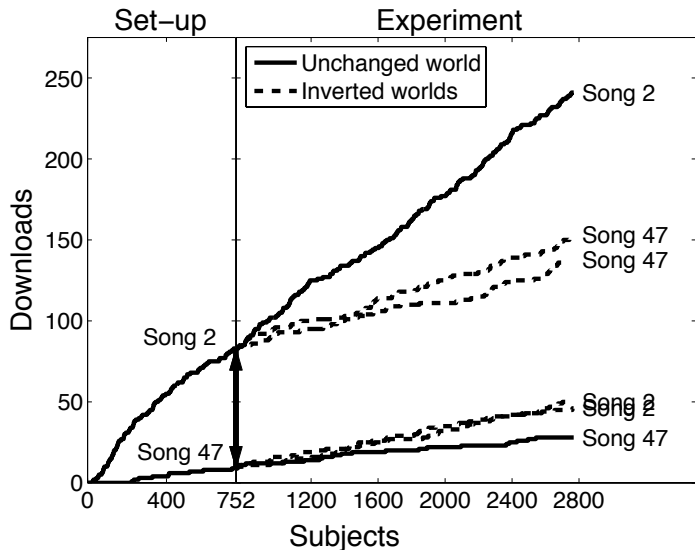
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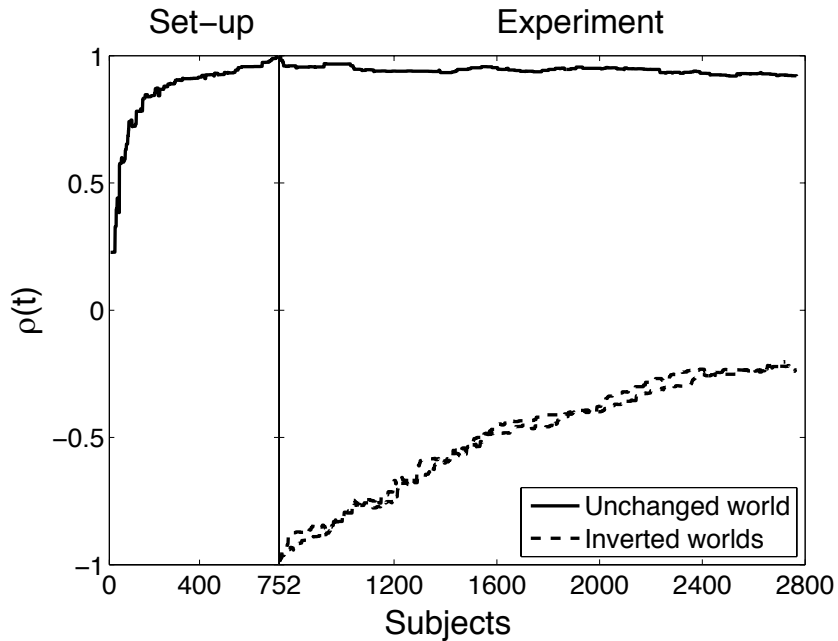


Tracking song 1 and song 48



Tracking song 2 and song 47









(Duchamp, 1919)



(Duchamp, 1919)



(Warhol, 1963)



(Duchamp, 1919)



(Warhol, 1963)



(Rohrer, 1999)



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- ▶ social influences weakens the relationship between appeal and success
- ▶ you can predict failure but you can't predict success
- ▶ the perception of success can become a self-fulfilling prophecy

Download all the data from these experiments:

<http://opr.princeton.edu/archive/cm/>

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Good luck on the midterm, and have a wonderful break!